习题

1-1 利用两矢量点积定义式

$$\because \quad \vec{A}\cdot\vec{B}=AB\cos\theta$$

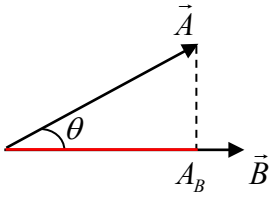
$$\cos\theta=\frac{\vec{A}\cdot\vec{B}}{\left|\vec{A}\right|\cdot\left|\vec{B}\right|}$$

$$\cos\theta=\frac{\left(\vec{e}_x+2\vec{e}_y+3\vec{e}_z\right)\bullet\left(4\vec{e}_x-5\vec{e}_y+6\vec{e}_z\right)}{\sqrt{1^2+2^2+3^2}\sqrt{4^2+5^2+6^2}}=\frac{4-10+18}{\sqrt{1078}}$$

$$=\frac{12}{\sqrt{1078}}$$

$$\therefore \quad \theta=\cos^{-1}\theta=68.56^\circ$$

$\vec{A}$ 在 $\vec{B}$ 上的分量如图



$$A_B=\vec{A}\bullet\vec{e}_B$$

$$A_B=\left(\vec{e}_x+2\vec{e}_y+3\vec{e}_z\right)\bullet\frac{\left(4\vec{e}_x-5\vec{e}_y+6\vec{e}_z\right)}{\sqrt{4^2+5^2+6^2}}$$

$$=\frac{4-10+18}{\sqrt{77}}=\frac{12}{\sqrt{77}}$$

1-2 给定两矢量 $\vec{A}=2\vec{e}_x+3\vec{e}_y-4\vec{e}_z$ 和 $\vec{B}=-6\vec{e}_x-4\vec{e}_y+\vec{e}_z$，求 $\vec{A}\times\vec{B}$ 在 $\vec{C}=\vec{e}_x-\vec{e}_y+\vec{e}_z$ 上的分量。

解：

$$\vec{A}\times\vec{B}=\begin{vmatrix}\vec{e}_x & \vec{e}_y & \vec{e}_z\\ 2 & 3 & -4\\ -6 & -4 & 1\end{vmatrix}=-13\vec{e}_x+22\vec{e}_y+10\vec{e}_z$$

$\vec{A}\times\vec{B}$ 在 $\vec{C}=\vec{e}_x-\vec{e}_y+\vec{e}_z$ 上的分量

$$\frac{(\vec{A}\times\vec{B})\bullet\vec{C}}{\left|\vec{C}\right|}=\frac{-13-22+10}{\sqrt{3}}=-14.43$$

1-3 已知 $f(x)=3r^2+4\ln r+\frac{6}{\sqrt[3]{r}}$，求 ∇f

解：应用复合微分求导的原则将位置矢量 $\boldsymbol{r}$ 看作中间变量

$$\nabla f = f'\nabla r = \left(3r^2 + 4\ln r + \frac{6}{\sqrt[3]{r}}\right)' \nabla r$$

$$= (6r + \frac{4}{r} - \frac{2}{\sqrt[3]{r^2}})\frac{\vec{r}}{r} = (6 + 4r^{-2} - 2r^{-\frac{7}{3}})\ \vec{r}$$

注：在直角坐标中：$\vec{r} = x\,\vec{e}_x + y\,\vec{e}_y + z\,\vec{e}_z$

1-4 求 $f(x,y,z) = x^2yz + 4xz^2$ 在 $P(1,-2,-1)$ 处沿 $\vec{A} = 2\vec{e}_x - \vec{e}_y - 2\vec{e}_z$ 方向的方向导数

解：$\nabla f = \frac{\partial f}{\partial x}\vec{e}_x + \frac{\partial f}{\partial y}\vec{e}_y + \frac{\partial f}{\partial z}\vec{e}_z = (2xyz + 4z^2)\vec{e}_x + x^2z\vec{e}_y + (x^2y + 8xz)\vec{e}_z$ （1）

又 $P(1,-2,-1)$，把 P 点代入式（1）得：

$$\nabla f\big|_p = \left(\frac{\partial f}{\partial x}\vec{e}_x + \frac{\partial f}{\partial y}\vec{e}_y + \frac{\partial f}{\partial z}\vec{e}_z\right)_p = \left[(2xyz + 4z^2)\vec{e}_x + x^2z\vec{e}_y + (x^2y + 8xz)\vec{e}_z\right]_p = 8\vec{e}_x - \vec{e}_y - 10\vec{e}_z$$

因为：$\vec{e}_A = \frac{\vec{A}}{|\vec{A}|} = \frac{2\vec{e}_x - \vec{e}_y - 2\vec{e}_z}{\sqrt{4+1+4}} = \frac{2\vec{e}_x - \vec{e}_y - 2\vec{e}_z}{3}$

所以：$\nabla f \cdot \vec{e}_A = (8\vec{e}_x - \vec{e}_y - 10\vec{e}_z) \cdot \frac{2\vec{e}_x - \vec{e}_y - 2\vec{e}_z}{3} = \frac{37}{3}$

1-5 试求空间曲面 $x^2y + 2xz = 4$ 在点 $P(2,-2,3)$ 处的法向单位矢量

解：将空间曲面看成是函数值为 4 的等值面，点 $P(2,-2,3)$ 在该曲面上，因此可设该函数为

$$f(x,y,z) = x^2y + 2xz$$

因为函数的梯度与等值面垂直，所以先求出该函数的梯度，则

$$\nabla f = \frac{\partial f}{\partial x}\vec{e}_x + \frac{\partial f}{\partial y}\vec{e}_y + \frac{\partial f}{\partial z}\vec{e}_z = (2xy + 2z)\vec{e}_x + x^2\vec{e}_y + 2x\vec{e}_z \qquad (1)$$

把 P 点代入式（1）得函数在该点处的梯度：

$$\nabla f\big|_p = \left(\frac{\partial f}{\partial x}\vec{e}_x + \frac{\partial f}{\partial y}\vec{e}_y + \frac{\partial f}{\partial z}\vec{e}_z\right)_p = \left[(2xy + 2z)\vec{e}_x + x^2\vec{e}_y + 2x\vec{e}_z\right]_p = -2\vec{e}_x + 4\vec{e}_y + 4\vec{e}_z$$

$$\therefore \vec{e}_n = \frac{-2\vec{e}_x + 4\vec{e}_y + 4\vec{e}_z}{\sqrt{4+16=16}} = \frac{-\vec{e}_x + 2\vec{e}_y + 2\vec{e}_z}{3}$$

1-6 已知 $\vec{F}(x,y,z) = x^2z\vec{e}_x - 2y^3z^2\vec{e}_y + xy^2z\vec{e}_z$，求点（1,-1,1）处的 $\nabla \bullet \vec{F}$

解：$\nabla \bullet \vec{F}\Big|_p = \left(\frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}\right)\Big|_p = \left(2xz - 6y^2z^2 + xy^2\right)_p = 2 - 6 + 1 = -3$

1-7 欲使 $\vec{F}(x,y,z) = (x-3y)\vec{e}_x + (y-2z)\vec{e}_y + (x+az)\vec{e}_z$ 的散度为零，试问常数 a 应为何值。

解：要使 $\nabla \cdot \vec{F} = 0$ 即 $\frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z} = 0$，又 $F_x = x - 3y, F_y = y - 2z, F_z = x + az$ 分别求偏导并代入得：$1 + 1 + a = 0$ 解得：$a = -2$

1-8 已知 $\vec{F}(x,y,z) = xz^3\vec{e}_x - 2x^2yz\vec{e}_y + 2yz^4\vec{e}_z$，求点（1,-1,1）处的 $\nabla \times \vec{F}$

解：首先求出函数 $\vec{F}$ 的旋度；

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix} = \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z}\right)\vec{e}_x + \left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x}\right)\vec{e}_y + \left(\frac{\partial F_y}{\partial x} + \frac{\partial F_x}{\partial y}\right)\vec{e}_z$$

$$= \left(2z^4 + 2x^2y\right)\vec{e}_x + \left(3xz^2 - 0\right)\vec{e}_y + \left(-4xyz - 0\right)\vec{e}_z$$

$$\nabla \times \vec{F}\Big|_{(1,-1,1)} = \left[\left(2xz^4 + 2x^2y\right)\vec{e}_x + \left(3xz^2 - 0\right)\vec{e}_y + \left(-4xyz - 0\right)\vec{e}_z\right]\Big|_{(1,-1,1)} = 3\vec{e}_y + 4\vec{e}_z$$

1-9 已知 $\vec{F}(x,y,z) = 3xy\vec{e}_x - y^2\vec{e}_y$，试求 $\vec{F}$ 沿曲线 $y = 2x^2$ 由点 $P_1(0,0)$ 至 $P_2(1,2)$ 的线积分

解：设 $x = t, y = 2t^2$，则 $dx = dt, dy = 4tdt$，$t \in [0,1]$，又：$d\vec{l} = dx\vec{e}_x + dy\vec{e}_y$

所以：$\vec{F} \bullet d\vec{l} = (3xy\vec{e}_x - y^2\vec{e}_y) \bullet (dx\vec{e}_x + dy\vec{e}_y) = 3xydx - y^2dy = (6t^3 - 16t^5)dt$

所以：$\int_{P_1}^{P_2} \vec{F} \bullet d\vec{l} = \int_0^1 (6t^3 - 16t^5)dt = \frac{3}{2}t^4 - \frac{8}{3}t^6\Big|_0^1 = -\frac{7}{6}$

1-10 已知 $f(x,y,z) = 2xyz^2$，$\vec{F}(x,y,z) = xy\vec{e}_x - z\vec{e}_y + x^2\vec{e}_z$，试就参数方程

$x=t^2, y=2t, z=t^3$（$0\le t\le 1$）的同一曲线 l，计算 t 由 0 变为 1 时的两个矢量线积分：

（1）$\int_l f d\vec{l}$

（2）$\int_l \vec{F}\times d\vec{l}$

解：（1）$\because x=t^2, y=2t, z=t^3$

$$\therefore \quad \mathrm{d}x=2tdt \ , \quad dy=2dt \ , \quad dz=t^2dt$$

$$\therefore \quad \int_l f d\vec{l}=\int_0^1 4t^9\left(2tdt\vec{e}_x+2dt\vec{e}_y+3t^2dt\vec{e}_z\right)=\int_0^1\left(8t^{10}\vec{e}_x+8t^9\vec{e}_y+12t^{11}\vec{e}_z\right)dt$$

（2）

$$=\left(\frac{8}{11}t^{11}\vec{e}_x+\frac{8}{10}t^{10}\vec{e}_y+t^{12}\vec{e}_z\right)\Big|_0^1=\frac{8}{11}\vec{e}_x+\frac{4}{5}\vec{e}_y+\vec{e}_z$$

$$\because \vec{F}=xy\vec{e}_x-z\vec{e}_y+x^2\vec{e}_z=2t^3\vec{e}_x-t^3\vec{e}_y+t^4\vec{e}_z$$

$$d\vec{l}=dx\vec{e}_x+dy\vec{e}_y=dz\vec{e}_z=2tdt\vec{e}_x+2dt\vec{e}_y+3t^2dt\vec{e}_z$$

$$\vec{F}\times d\vec{l}=\begin{vmatrix}\vec{e}_x & \vec{e}_y & \vec{e}_z\\ F_x & F_y & F_z\\ dx & dy & dz\end{vmatrix}=(-3t^5-2t^4)dt\vec{e}_x-4t^5dt\vec{e}_y+(4t^3dt+2t^4dt)\vec{e}_z$$

$$\therefore \quad \int_l \vec{F}\times d\vec{l}=\int_0^1\left[(-3t^5-2t^4)\,\vec{e}_x-4t^5\,\vec{e}_y+(4t^3+2t^4)\vec{e}_z\right]dt$$

$$=\left[(-\frac{1}{2}t^6-\frac{2}{5}t^5)\,\vec{e}_x-\frac{2}{3}t^5\,\vec{e}_y+(t^4+\frac{2}{5}t^5)\vec{e}_z\right]_0^1=-\frac{9}{10}\vec{e}_x-\frac{2}{3}\vec{e}_y+\frac{7}{5}\vec{e}_z$$

1-11 对于$f(\boldsymbol{r})$和$\boldsymbol{F}(\boldsymbol{r})$，证明下列两恒等式：

（1）$\nabla\times(f\boldsymbol{F})=f(\nabla\times\boldsymbol{F})+\nabla f\times\boldsymbol{F}$

由旋度公式展开

$$\nabla\times\vec{F}=\begin{vmatrix}\vec{e}_x & \vec{e}_y & \vec{e}_z\\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z}\\ F_x & F_y & F_z\end{vmatrix}=(\frac{\partial F_z}{\partial y}-\frac{\partial F_y}{\partial z})\vec{e}_x+(\frac{\partial F_x}{\partial z}-\frac{\partial F_z}{\partial x})\vec{e}_y+(\frac{\partial F_y}{\partial x}+\frac{\partial F_x}{\partial y})\vec{e}_z$$

$$\nabla\times(f\vec{F})=\begin{vmatrix}\vec{e}_x & \vec{e}_y & \vec{e}_z\\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z}\\ f\overrightarrow{F_x} & f\overrightarrow{F_y} & f\overrightarrow{F_z}\end{vmatrix}$$

$$=\left[\frac{\partial(f\overrightarrow{Fz})}{\partial y}-\frac{\partial(f\overrightarrow{Fy})}{\partial z}\right]\overrightarrow{e_x}+\left[\frac{\partial(f\overrightarrow{Fx})}{\partial z}-\frac{\partial(f\overrightarrow{Fz})}{\partial x}\right]\overrightarrow{e_y}+\left[\frac{\partial(f\overrightarrow{Fy})}{\partial x}-\frac{\partial(f\overrightarrow{Fx})}{\partial y}\right]\overrightarrow{e_z}$$

由于，$\dfrac{\partial(f\overrightarrow{Fy})}{\partial z}=\dfrac{\partial f}{\partial z}\cdot\overrightarrow{Fy}+f\cdot\dfrac{\partial\overrightarrow{Fy}}{\partial z}$，其他同理代入后得到

$$\nabla\times(f\vec{F})=f(\frac{\partial\overrightarrow{F_z}}{\partial y}-\frac{\partial\overrightarrow{F_y}}{\partial z})\vec{e}_x+f(\frac{\partial\overrightarrow{F_x}}{\partial z}-\frac{\partial\overrightarrow{F_z}}{\partial x})\vec{e}_y+f(\frac{\partial\overrightarrow{F_y}}{\partial x}+\frac{\partial\overrightarrow{F_x}}{\partial y})\vec{e}_z$$

$$+(\frac{\partial f}{\partial y}\cdot\overrightarrow{Fz}-\frac{\partial f}{\partial z}\cdot\overrightarrow{Fy})\vec{e}_x+(\frac{\partial f}{\partial z}\cdot\overrightarrow{Fx}-\frac{\partial f}{\partial x}\cdot\overrightarrow{Fz})\vec{e}_y+(\frac{\partial f}{\partial x}\cdot\overrightarrow{Fy}-\frac{\partial f}{\partial y}\cdot\overrightarrow{Fx})\vec{e}_z$$

因为 $\nabla f=\dfrac{\partial f}{\partial x}\vec{e}_x+\dfrac{\partial f}{\partial y}\vec{e}_y+\dfrac{\partial f}{\partial z}\vec{e}_z$

所以可以化简为

$$\nabla\times(f\vec{F})=f\begin{vmatrix}\vec{e}_x & \vec{e}_y & \vec{e}_z\\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z}\\ \overrightarrow{F_x} & \overrightarrow{F_y} & \overrightarrow{F_z}\end{vmatrix}+\begin{vmatrix}\vec{e}_x & \vec{e}_y & \vec{e}_z\\ \dfrac{\partial f}{\partial x} & \dfrac{\partial f}{\partial y} & \dfrac{\partial f}{\partial z}\\ \overrightarrow{F_x} & \overrightarrow{F_y} & \overrightarrow{F_z}\end{vmatrix}$$

$=f\nabla\times\vec{F}+\nabla f\times\vec{F}$

（2）$(\boldsymbol{F}\cdot\nabla)f=\boldsymbol{F}\cdot\nabla f$

已知，

$$\nabla=\frac{\partial}{\partial x}\vec{e}_x+\frac{\partial}{\partial y}\vec{e}_y+\frac{\partial}{\partial z}\vec{e}_z$$

$$(\vec{F}\cdot\nabla)=\overrightarrow{F_x}\frac{\partial}{\partial x}\vec{e}_x+\overrightarrow{F_y}\frac{\partial}{\partial y}\vec{e}_y+\overrightarrow{F_z}\frac{\partial}{\partial z}\vec{e}_z$$

$$(\vec{F}\cdot\nabla)f=\overrightarrow{F_x}\frac{\partial f}{\partial x}\vec{e}_x+\overrightarrow{F_y}\frac{\partial f}{\partial y}\vec{e}_y+\overrightarrow{F_z}\frac{\partial f}{\partial z}\vec{e}_z$$

又因为，$\nabla f=\dfrac{\partial f}{\partial x}\vec{e}_x+\dfrac{\partial f}{\partial y}\vec{e}_y+\dfrac{\partial f}{\partial z}\vec{e}_z$

故$(\vec{F}\cdot\nabla)f=\overrightarrow{F_x}\dfrac{\partial f}{\partial x}\vec{e}_x+\overrightarrow{F_y}\dfrac{\partial f}{\partial y}\vec{e}_y+\overrightarrow{F_z}\dfrac{\partial f}{\partial z}\vec{e}_z=\vec{F}\cdot\nabla f$

1-12 对于标量场 $f\left(\vec{r}\right)$ 和 $g\left(\vec{r}\right)$，试证明：

（1） $\nabla\times\left(f\nabla f\right)=0$

（2） $\nabla^2\left(fg\right)=f\nabla^2 g+g\nabla^2 f+2\nabla f\bullet\nabla g$

证明：（1）$\because\nabla\times\left(f\nabla f\right)=\nabla f\times\nabla f+f\nabla\times\left(\nabla f\right)$，又 $\nabla f\times\nabla f=0,\nabla\times\left(\nabla f\right)=0$。故左式=0 得到证明。

证明（2） 左边

$$\because\nabla^2\left(fg\right)=\nabla\bullet\nabla\left(fg\right)$$

$$\nabla\left(fg\right)=f\nabla g+g\nabla f$$

$$\nabla\bullet\left(f\nabla g+g\nabla f\right)=\nabla f\bullet\nabla g+f\nabla^2 g+\nabla g\bullet\nabla f+g\nabla^2 f$$
$$=f\nabla^2 g+g\nabla^2 f+2\nabla g\bullet\nabla f$$

所以左式=右式 原题得到证明。

1-13 试计算

（1）$\nabla^2\ln r$

解：

$$\nabla^2\ln r=\nabla\bullet\nabla\ln r=\nabla\bullet\left(\frac{1}{r}\nabla r\right)=\left(\frac{1}{r}\cdot\frac{\vec{r}}{r}\right)=\nabla\bullet\left(\frac{\vec{r}}{r^2}\right)$$

$$=\frac{1}{r^2}\nabla\bullet\vec{r}+\nabla\frac{1}{r^2}\bullet\vec{r}=\frac{3}{r^2}+(-2)r^{-4}\vec{r}\bullet\vec{r}=r^{-2}$$

（2）$\nabla^2[\nabla\bullet(\frac{\vec{r}}{r^2})]$

解：利用上题的结果 $\nabla\bullet\left(\frac{\vec{r}}{r^2}\right)=r^{-2}$，因此

$$\nabla^2\left[\nabla\bullet\left(\frac{\vec{r}}{r^2}\right)\right]=\nabla^2\left(r^{-2}\right)=\nabla\bullet\nabla r^{-2}=\nabla\bullet\left(\frac{2\vec{r}}{r^4}\right)=-2\left[\frac{1}{r^4}\nabla\bullet\vec{r}+\nabla\frac{1}{r^4}\bullet\vec{r}\right]$$

$$=-2\left[\frac{3}{r^4}+\frac{-4}{r^5}\frac{\vec{r}}{r}\bullet\vec{r}\right]=2r^{-4}$$

1-14 对于平面矢量场 $\boldsymbol{F}(x,y,z)=F_x(x,y)\boldsymbol{e}_x+F_y(x,y)\boldsymbol{e}_y$ 和 xoy 平面上沿逆时针取向的闭面路径 l 及其所围区域 S，试由斯托克斯定理导出下列所谓平面格林定理：

$$\oint_L \overrightarrow{Fy}\cdot dy+\overrightarrow{Fx}\cdot dx=\iint_S\left(\frac{\partial\overrightarrow{Fy}}{\partial x}-\frac{\partial\overrightarrow{F_x}}{\partial y}\right)dxdy$$

解：由 $\int\overrightarrow{Fx}\cdot dx+\overrightarrow{Fy}\cdot dy=\int\vec{F}\cdot d\vec{l}$，

再由斯托克斯定理，$\int\vec{F}\cdot d\vec{l}=\int(\nabla\times\vec{F})\cdot d\vec{S}$

旋度公式，且对于平面，则有 $\overrightarrow{F_z}=0$；

$$\nabla\times\vec{F}=\begin{vmatrix}\vec{e}_x & \vec{e}_y & \vec{e}_z\\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z}\\ F_x & F_y & 0\end{vmatrix}=\left(-\frac{\partial\overrightarrow{F_y}}{\partial z}\right)\vec{e}_x+\left(\frac{\partial\overrightarrow{F_x}}{\partial z}\right)\vec{e}_y+\left(\frac{\partial\overrightarrow{Fy}}{\partial x}-\frac{\partial\overrightarrow{F_x}}{\partial y}\right)\vec{e}_z$$

由于 $d\vec{S}=dxdy\vec{e}_z$ 代入得到，$\oint_L\overrightarrow{Fy}\cdot dy+\overrightarrow{Fx}\cdot dx=\iint_S\left(\frac{\partial\overrightarrow{Fy}}{\partial x}-\frac{\partial\overrightarrow{F_x}}{\partial y}\right)dxdy$

1-15 判断下列两矢量场各自属于哪种类型：

（1）$\vec{\mathrm{F}}=(6xy+z^3)\vec{e}_x+(3x^2-z)\vec{e}_y+(3xz^2-y)\vec{e}_z$

（2）$\vec{G}=3y^4z^2\vec{e}_x+x^3z^2\vec{e}_y-3x^2y^2\vec{e}_z$

解：根据判据 $\nabla\bullet\vec{F}=0$ 和 $\nabla\times\vec{F}=0$ 是否成立可以判断

（1）式是无旋有散场

（2）式是有旋无散场

1-16 设 F(r)和 G(r)均为无旋场，试证明：

$$\nabla\cdot(F\times G)=0$$

因为是无旋场所以 $\begin{matrix}\nabla\times F=0\\ \nabla\times G=0\end{matrix}$，

公式展开有：$\nabla\cdot(F\times G)=G\cdot(\nabla\times F)+F\cdot(\nabla\times G)=0+0=0$

1-17 已知 $\nabla\phi=\left(y^2-2xyz^3\right)\vec{e}_x+\left(3+2xy-x^2z^3\right)\vec{e}_y+\left(6z^3-3x^2yz^2\right)\vec{e}_z$，试求 $\varphi(x,y,z)$。

解：令 $\frac{\partial\varphi}{\partial x}=y^2-2xyz^3$， 积分得：

$\varphi = xy^2 - x^2yz^3 + f\left(y,z\right)$，

由：$\dfrac{\partial\varphi}{\partial y} = \dfrac{\partial\left(xy^2 - x^2yz^3 + f\left(y,z\right)\right)}{\partial y} = 3 + 2xy - x^2z^3$

从而积分解得：$f\left(y,z\right) = 3y + g(z)$

则 $\varphi = xy^2 - x^2yz^3 + 3y + g(z)$

由：$\dfrac{\partial\varphi}{\partial z} = \dfrac{\partial[xy^2 - x^2yz^3 + 3y + g(z)]}{\partial z} = 6z^3 - 3x^2yz^2$

从而解得：$g(z) = \dfrac{3}{2}z^4 + C$

由以上可得：$\varphi\left(x,y,z\right) = xy^2 + 3y - x^2yz^3 + 1.5z^4 + c$